

Can an Unpaired Sliced-Wasserstein Autoencoder Learn the Detector Response?

Identification, Zero-Shot Transfer to the Υ Family, and Per-Event Unfolding on CMS Open Data

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Abstract

Unpaired transport models such as the sliced-Wasserstein autoencoder (SWAE) learn a stochastic map between truth-level and detector-level four-vectors from marginals alone. Marginal agreement, however, does not establish that the model has learned the response: a deterministic map can absorb the detector resolution into its mean and still match every one-dimensional projection. We train a single mass-blind SWAE on real CMS 2012 open data together with a theory prior for the J/ψ and Z resonances, and ask three questions. First, is the response identifiable - does the learned stochastic channel carry the resolution? Second, does the same object transfer to the $\Upsilon(1S, 2S, 3S)$ family, which is excluded from training? Third, does it invert per event? We answer the first with a fixed-truth repeated-decoding budget and a scorecard that reports every gauge against its identity and finite-sample floor, and we show that noise-aware checkpoint selection picks the deterministic solution in eight of eight arms. Fixing the noise amplitude from an independent resolution measurement restores the channel: the $\Upsilon(1S)$ intra-event width becomes 86.5-91.6 MeV against the 84 MeV CMS resolution. The same model predicts the Υ peak positions to +21.6/ + 2.9/ + 6.3 MeV with a best-to-last drift of 1-3 MeV and a systematic budget dominated by the checkpoint choice (85-92 MeV), while the peak widths remain over-spread (290-299 MeV) and per-event unfolding improves on the identity map but does not reach the oracle. We report the refuted mechanisms (a stronger learned noise amplitude, a cycle-consistent noise schedule, a heavier per-event weight) as constraints, and we state explicitly what a claim of learned detector response requires. **[TODO: Final abstract numbers after the open arms settle.]**

Introduction

Fast, data-driven detector simulation and unfolding are central applications of machine learning in high-energy physics^{1;2}. The OTUS/SWAE formulation^{1;3;4} learns an unpaired transport between a physically interpretable truth space $\mathcal{Z}$ and detector-level observations $\mathcal{X}$ with a sliced-Wasserstein objective,

$$\mathcal{L} = \mathbb{E}_x \mathbb{E}_{z \sim E(x)} \mathbb{E}_{\tilde{x} \sim D(z)} \|x - \tilde{x}\|^2 + \lambda d_{\text{SW}}(p_E(z), p(z)), \quad (1)$$

where E is the encoder (unfolding) and D the decoder (simulation).

The literal objective is a *marginal* criterion. Our starting observation, which motivates this paper, is that marginal agreement is not evidence of a learned response: because the latent ensemble varies in coordinates correlated with the resonance mass, a deterministic decoder

can reproduce the observed mass spread while its stochastic channel collapses, and every one-dimensional projection still closes. We turn that observation into three questions and answer each with a measurement rather than a metric:

- **Identification (RQ1).** Does the stochastic channel carry the detector resolution, or is the resolution manufactured by the mean map?
- **Transfer (RQ2).** Does one shared, mass-blind response trained on J/ψ and Z predict a resonance that never entered training - the $\Upsilon(1S, 2S, 3S)$ family - in both peak position and width?
- **Inversion (RQ3).** Does the same trained object unfold per event, with calibrated uncertainty, where paired truth exists?

Related work places this in a crowded landscape. For unfolding, OmniFold is the current reference⁵, and discriminative reweighting without iterations (AUSSIE, arXiv:2602.24282) and Wasserstein-loss unfolding with an explicit injectivity condition (arXiv:2603.20903) are the closest modern alternatives; both are used here as external reference points rather than as baselines we claim to beat. The immediate predecessor of this work is the OTUS autoencoder¹, which established that an unpaired transport can be learned between theory-level and detector-level samples without any (z, x) pairs. Its two demonstrations, however, generated **both** sides with MadGraph5, Pythia8 and Delphes: the detector it learned was itself a simulation, so the response could be checked against the generator that produced it. This paper replaces that simulated detector level with real CMS open data and therefore has no such oracle. CWola inverted detector effects with conditional generative networks⁶, and the unpaired line continues with point-cloud translation⁷. On the simulation side, detector-level generative modelling has moved from GANs^{2;8} to flows, diffusion and particle-cloud models⁹; the neural-optimal-transport line^{10;11} and flow matching^{12;13} supply the building blocks we reuse. What is missing across that literature, and what this paper supplies, is a criterion for deciding whether an unpaired model has learned the response at all.

The work makes three moves. (i) **Real detector data.** The detector level is CMS 2012 open data rather than a fast simulation, so the learned response is the real detector’s and there is no generator to validate against. (ii) **A different network.** Encoder and decoder are cylindrical-coordinate residual flows with an explicit per-event noise channel (a Gaussian core and a Student- t tail), conditioned on muon kinematics only, and trained jointly on J/ψ and Z with one shared, mass-blind parameter set. (iii) **A criterion and a measurement.** An identification scorecard that asks whether the resolution is carried by the noise or faked by the mean map, the demonstration that noise-aware marginal selection is actively harmful (eight of eight arms), a frozen-amplitude calibration and what it buys, a zero-shot transfer test to the Υ family that separates what transfers (peak positions) from what does not (widths, conditional shape), and a quantified systematics budget with the refuted mechanisms. **[TODO: Sharpen the contribution list after the open arms settle.]**

Table 1: **In-domain closure.** Joint Run E, selected checkpoint: mass and pair- p_T gauges [measured]. Identity and floor columns are still to be quoted per row.

region	direction	mass W_1 [GeV]	mass KS	pair- p_T KS
J/ψ	$z \rightarrow x$	0.00088	0.017	0.052
J/ψ	$x \rightarrow z$	0.00177	0.037	0.022
Z	$z \rightarrow x$	0.086	0.013	0.006
Z	$x \rightarrow z$	0.053	0.007	0.005

Table 2: **Identification scorecard.** Every arm from Run E to the Run H family, selected checkpoints [measured]. Ratios are intra-event width over the required resolution; degenerate means ratio below 0.5.

arm (selected checkpoint)	J/ψ ratio	Z ratio	slice ratio	key
Run E (last)	1.68	0.088	7.2	failed
Run F (last)	1.82	0.038	9.1	failed
Run G tf32 (last)	1.84	0.043	13.8	failed
Run H (last)	1.73	0.038	6.9	failed
Run H fix (last)	1.80	0.042	8.9	failed
A1 cycle (last)	1.74	0.038	9.7	failed
A2.3 frozen (best)	1.087	0.829	44.6	identified
A2.2 floor (best)	1.22	0.849	75.1	identified, correction rejected

Results

In-domain closure of a shared $J/\psi + Z$ response

One model closes both resonances in the marginals. With the selected checkpoint the mass and pair- p_T gauges are at or near their finite-sample floors (Table 1), and the same checkpoint is competitive on the continuum-reweighted evaluation prior, so the closure is not an artefact of a single prior.

The stochastic channel is not identified by marginal objectives

With a learned noise amplitude the channel collapses: the tail scale falls from 8.1×10^{-4} to 1.75×10^{-4} and the core scale sits on its floor, while the late-training Υ shift ($-125/-136/-136$ MeV) is unchanged when the noise multipliers are set to 0, 0.25, 0.5 or 1.0, and unchanged with all noise off. The resolution therefore lives in the deterministic mean map. Scoring validation at the checkpoint’s native noise - the intuitive remedy - selects the deterministic stage-one warmup in eight of eight arms, because the score is a marginal gauge and a stochastic checkpoint scores worse at its own operating point.

Freezing the amplitude to a kernel fitted to the measured resolution restores the channel: the $\Upsilon(1S)$ intra-event width is 86.5-91.6 MeV against the 84 MeV CMS resolution, with $\sigma_{\text{only}} = 91-101$ MeV, where the ungated arms give 10.5-11.0 MeV. Table 2 summarises the resulting scorecard, whose selection key rejects degenerate checkpoints before ranking by conditional closure.

Table 3: Υ **transfer**. Continuum-reweighted prior, frozen-kernel checkpoints [measured]: median minus the CMS fit and the decoded $1S$ width.

checkpoint	$1S$	$2S$	$3S$	$1S$ width [MeV]
stage-3 best (g110)	+21.6	+2.9	+6.3	299
last (g180)	+18.9	+6.9	+13.3	291

Table 4: **Systematics budget**. Spread of each claim across the prior, checkpoint and evaluation-seed axes, with the bootstrap statistical floor [measured].

claim	value	prior	checkpoint	seed
$1S$ median – CMS [MeV]	+20.9	0.06	84.8	1.09
$1S$ width [MeV]	298	0.9	35.6	1.4
in-domain J/ψ ratio	1.083	n/a	n/a	0.084
in-domain Z ratio	0.848	n/a	n/a	0.028
J/ψ slice ratio	38.3	n/a	n/a	7.9

Zero-shot transfer to the Υ family

Peak positions transfer, with a best-to-last drift of 1-3 MeV; the widths do not - 290 to 299 MeV against the 84 MeV resolution - and the conditional response remains wrong at fixed pair- p_T (per-slice W_1 /floor of 44-87). Table 4 quantifies what each claim depends on.

The peak-position claim is prior-robust (the raw and continuum-reweighted priors move it by 0.06 MeV) and seed-robust (1.1 MeV, with a bootstrap floor of 0.4 MeV) but checkpoint-fragile: the retained checkpoints of the same arm span 85 MeV. The selection rule, not the evaluation noise, is the dominant systematic. The prior does move the continuum - the 8.5-9.25 GeV over CMS density ratio changes from 1.57 to 0.948 for the same checkpoint - while leaving the peak where it is.

Per-event unfolding on a paired benchmark

The encoder removes the global mass bias but leaves the per-event scatter essentially unchanged, and the posterior is not calibrated (pull standard deviation 12.6-20.7 against a nominal 1; 1σ coverage 0.06-0.10 against a nominal 0.683).

Discussion

Marginal closure is necessary but not sufficient; the identification scorecard is our proposal for what to report instead, and the systematics budget shows how to read a transfer claim once the selection rule is declared. The response is demonstrated for a fixed two-particle final state, no theory or detector systematic uncertainty is propagated, and the refuted mechanisms (a stronger learned noise amplitude, a cycle-consistent noise schedule, a heavier per-event weight) constrain the explanation without yet providing the missing ingredient: an objective that ties the resolution to the noise channel while leaving the mean map identifiable. **[TODO: Quantify the training-seed axis (needs new runs).]**

Table 5: **Paired closure.** Benchmark with paired truth, 160k held-out pairs [measured].

map	residual rms vs identity
identity (hand back the detector event)	1.0000
upstream OTUS encoder	3.3304
this work, stage-1 checkpoint	0.9808
this work, stage-2 checkpoint	0.9843
oracle (best correction fitted on the pairing)	0.8099

Methods

Data and selection

Detector-level data are CMS 2012 open-data opposite-sign dimuons, selected in the J/ψ and Z windows; truth-level events are generated with MadGraph5 + Pythia8 and passed through the same fiducial selection. The two samples are never paired event-by-event. The Υ sample is held out: it is never used for training, tuning, checkpoint selection or metric design, and is opened only after the model is frozen. Locked splits [measured]: the legacy-prior group provides 2,899,563 detector-level and 71,659 truth-level J/ψ events, and 4,207,696 detector-level and 800,000 truth-level Z events; the honest component-aware group provides 501,317 truth-level J/ψ and 401,718 truth-level Z events over the same detector-level stream. **[TODO: Selection tables, split manifest, cache fingerprints.]**

Priors

The J/ψ and Z priors are component-aware mixtures (signal plus continuum). Two regimes exist and must be quoted with every in-domain number: a legacy hand-smeared J/ψ prior, whose required additional smearing is 12.2 MeV, and the honest component-aware prior, whose requirement is 30.0 MeV - a factor 2.46. A sideband-derived continuum-reweighted Υ prior is the primary transfer readout.

Model

Encoder and decoder are short residual flows in cylindrical coordinates, sharing one mass-blind parameter set; at each of K steps a physics-conditioned network predicts a bounded mean residual, a Gaussian core scale and a Student- t tail scale, with azimuthal transport on S^1 and energies reconstructed on the muon mass shell. The response is written as $x = M(z) + \epsilon$ with $\epsilon \sim p_\theta(\epsilon | z)$. The trained joint arms carry 830,710 parameters (415,355 per direction) in four hidden layers of 256 units with two flow steps, i.e. 3.32 MB of float32 weights [measured].

What changed relative to the predecessor

Three changes distinguish this model from the OTUS autoencoder¹, all source-verified in the repository:

- **Data.** The detector level is CMS 2012 open data, not a Delphes sample, so the response has a real, irreducible resolution and no oracle.

- **Network.** Instead of a plain multilayer autoencoder, both maps are short residual flows in cylindrical coordinates: per step a physics-conditioned network predicts a bounded mean residual and the two noise scales, azimuth is transported on S^1 , and energies are reconstructed on the muon mass shell.
- **Conditioning and scope.** The maps are conditioned on muon kinematics only (the explicit pair-mass coordinate is excluded and the exclusion is enforced by a contract test), and one shared parameter set is trained jointly on J/ψ and Z with balanced per-region statistics.

[TODO: Add the predecessor’s own architecture numbers side by side once the comparison run is in place.]

Objective

The loss combines the paired reconstruction cycle $D(E(x))$, a latent prior term on $E(x)$, a direct generator term $D(z)$, a p_T -sliced conditional mass term, and a z-space consistency cycle

$$\mathcal{L}_z = \|\text{std}(z) - \text{std}(E(D(z)))\|^2, \quad (2)$$

which asks the composition of the two maps to be the identity on the truth side; the x-side cycle alone can be satisfied by a deterministic autoencoder.

Identification diagnostics

For a fixed truth event we decode repeatedly and measure the intra-event width; the exact split $\text{Var}(x) = \mathbb{E}_z[\text{Var}(x | z)] + \text{Var}_z[\mathbb{E}(x | z)]$ separates the noise share from the mean-map share. The response gate requires the intra-event width to lie within 0.8-1.25 of the required resolution and $\sigma_{\text{only}} \geq 50$ MeV on the Υ states; a checkpoint is flagged degenerate when the ratio falls below 0.5. Conditional closure is scored per equal-count pair- p_T slice as W_1 over its finite-sample floor.

Checkpoint selection

Selection is a tiered rule rather than a weighted sum: reject any checkpoint whose response gate fails or that is degenerate, rank the survivors by conditional closure and then by the in-domain score, and report identification failure explicitly rather than crowning a marginal winner. The transfer axis is report-only and provably cannot enter the key.

Systematics and statistics

Four axes are quantified for each claim: the evaluation prior, the checkpoint, the subsample seed, and a bootstrap of the decoded sample. The reported spreads are the half-ranges of each axis; the bootstrap is the statistical floor, not a systematic. A training-seed axis requires new runs and is not yet measured. [TODO: Add the training-seed axis and the external comparison (AUSSIE on the paired benchmark).]

Data availability

CMS 2012 open data are public; the selection, the truth priors and every run directory are listed in the repository record. **[TODO: DOI and repository link.]**

Code availability

All training, diagnostic and evaluation code is in the repository, with 443 unit tests at the time of writing. **[TODO: Zenodo DOI for the exact commit.]**

Acknowledgements

[TODO:]

Author contributions

[TODO:]

Competing interests

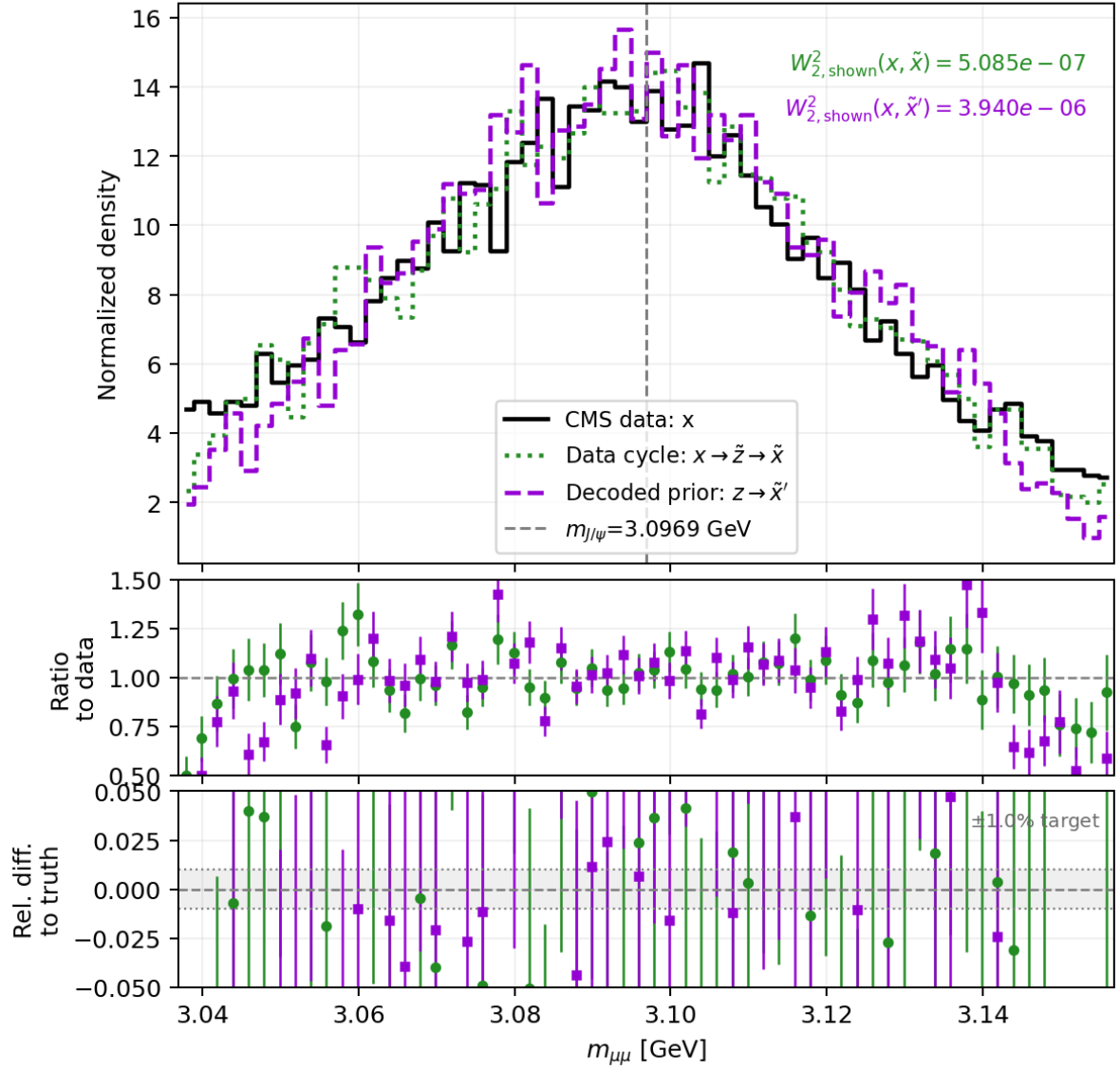
The authors declare no competing interests.

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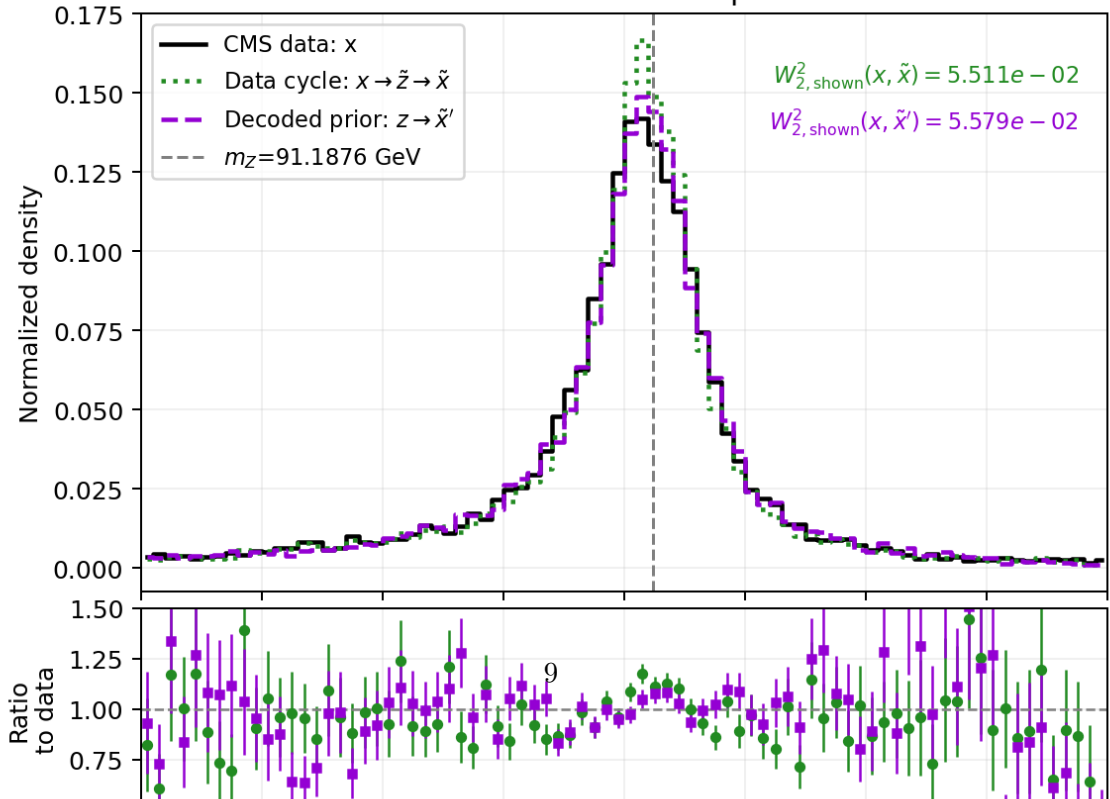
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Run E — $J/\psi \rightarrow \mu\mu$: x-space mass



Run E — $Z \rightarrow \mu\mu$: x-space mass



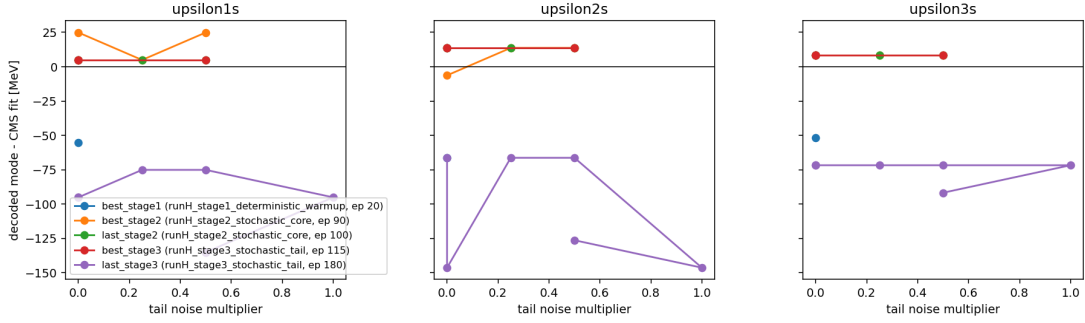


Figure 2: **The late Υ shift is not noise.** Decoded Υ peak position against the inference-time tail-noise multiplier, for healthy checkpoints and for the broken one; the broken checkpoint shows the full shift with the tail noise switched off. [measured]

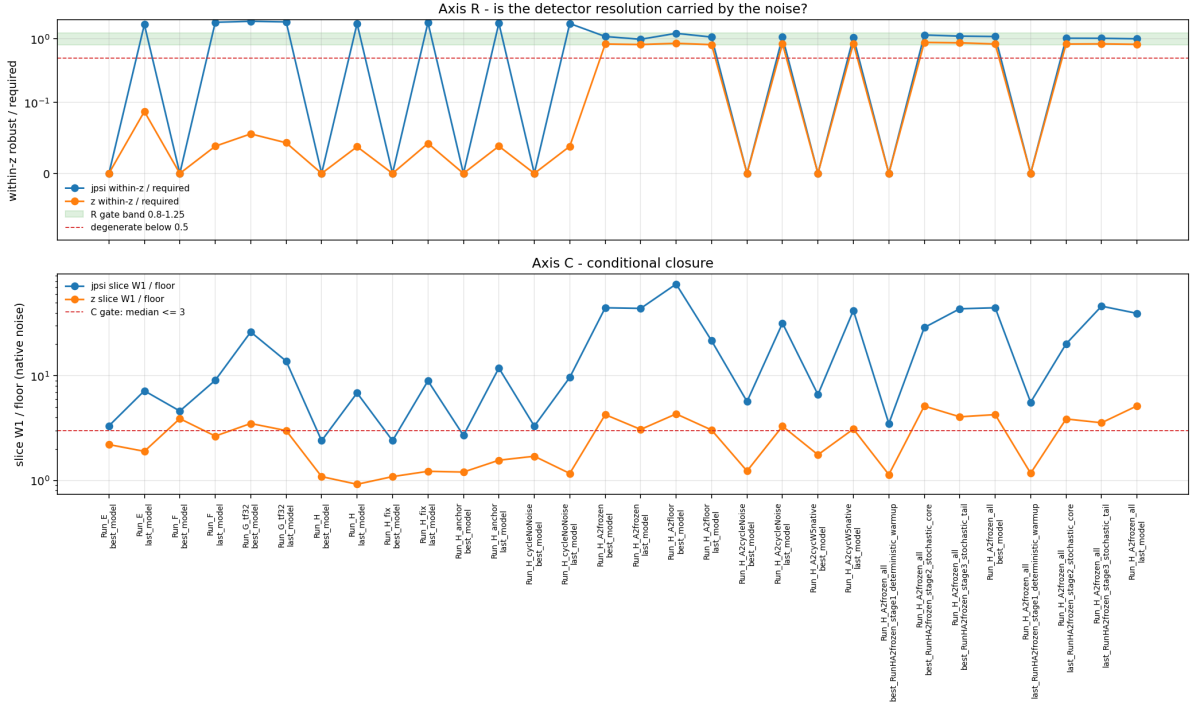


Figure 3: **Identification scorecard.** Top: intra-event width over the required resolution for every retained checkpoint of the Run E to Run H arms (band 0.8-1.25; degenerate below 0.5). Bottom: per-slice W_1 over its finite-sample floor at native noise. [measured]

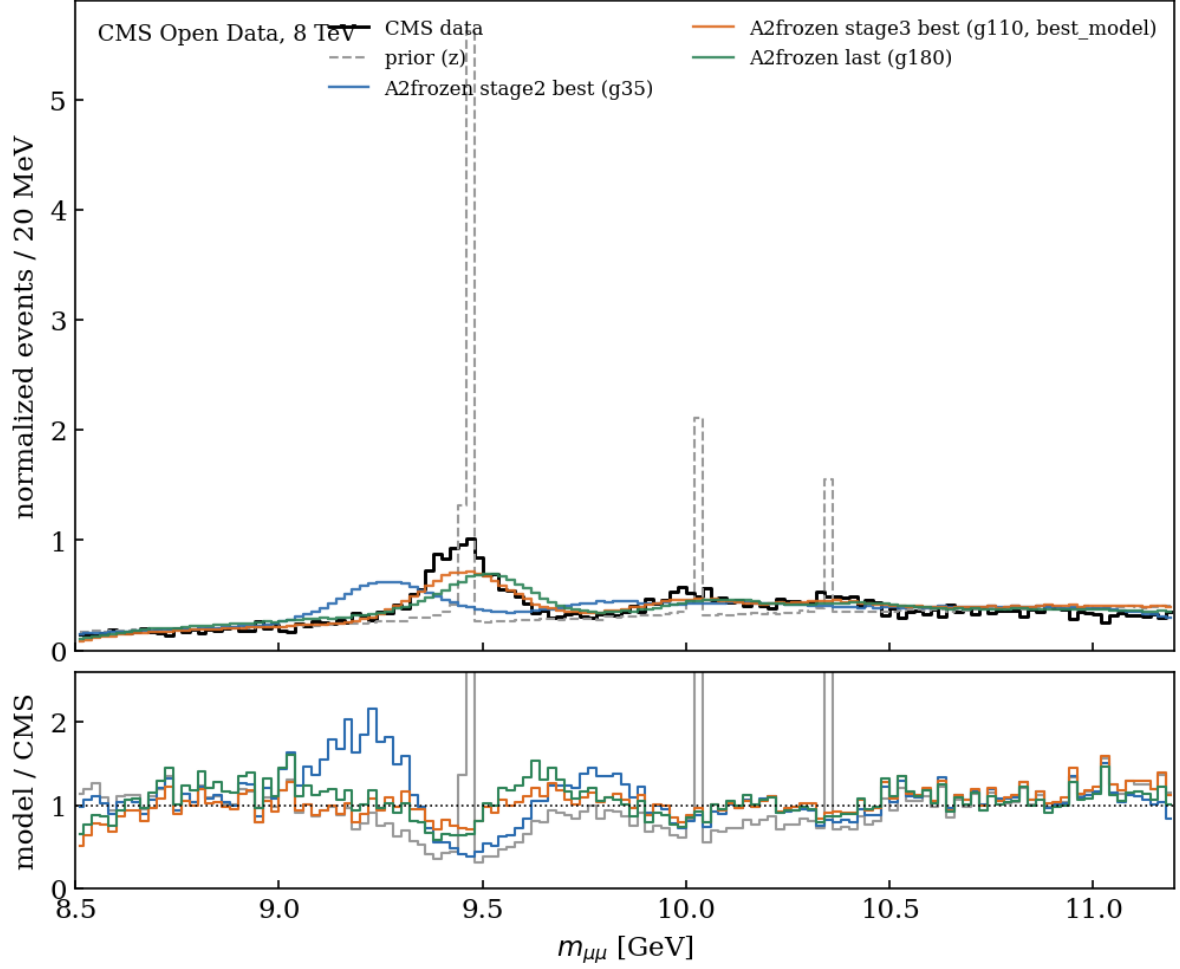


Figure 4: **Zero-shot transfer to the Υ family.** Decoded spectra on the continuum-reweighted evaluation prior against CMS data for three frozen-kernel checkpoints; the lower panel is the ratio to the data. Peak positions agree, the widths do not. [measured]

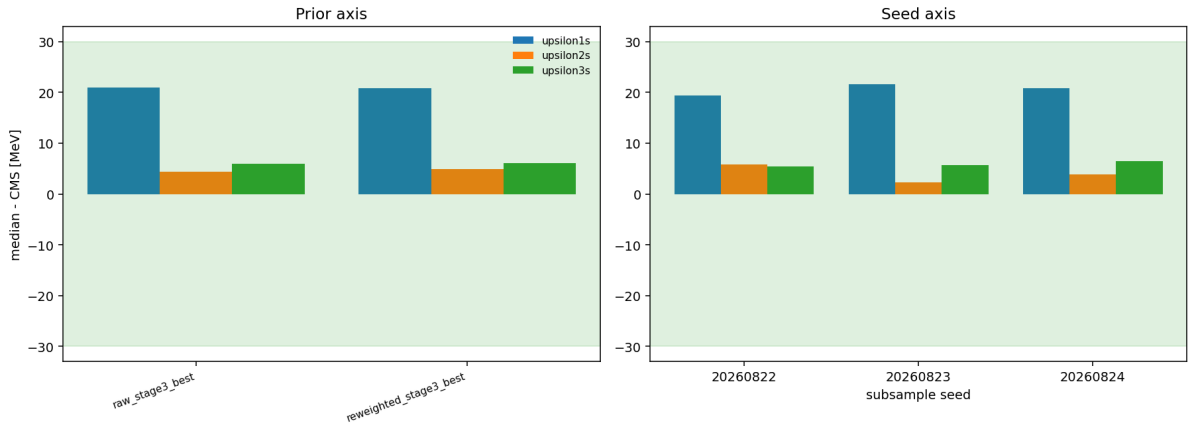


Figure 5: **Systematics budget.** Left: per-state medians on the raw and continuum-reweighted evaluation priors against the ± 30 MeV band. Right: the same quantity across three subsample seeds (the bootstrap floor is 0.4 MeV). The third axis - the choice of checkpoint - spans 85-92 MeV. [measured]

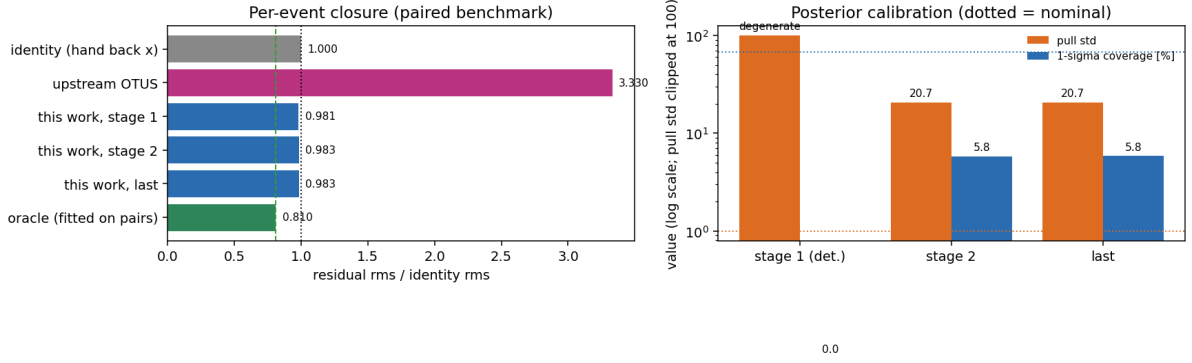


Figure 6: **Per-event unfolding.** Left: residual rms relative to the identity map on the paired benchmark; the oracle is fitted on the pairing and is a ceiling, not a method. Right: pull standard deviation and 1σ coverage against their nominal values; the deterministic stage-1 encoder has a zero-variance posterior. [measured]

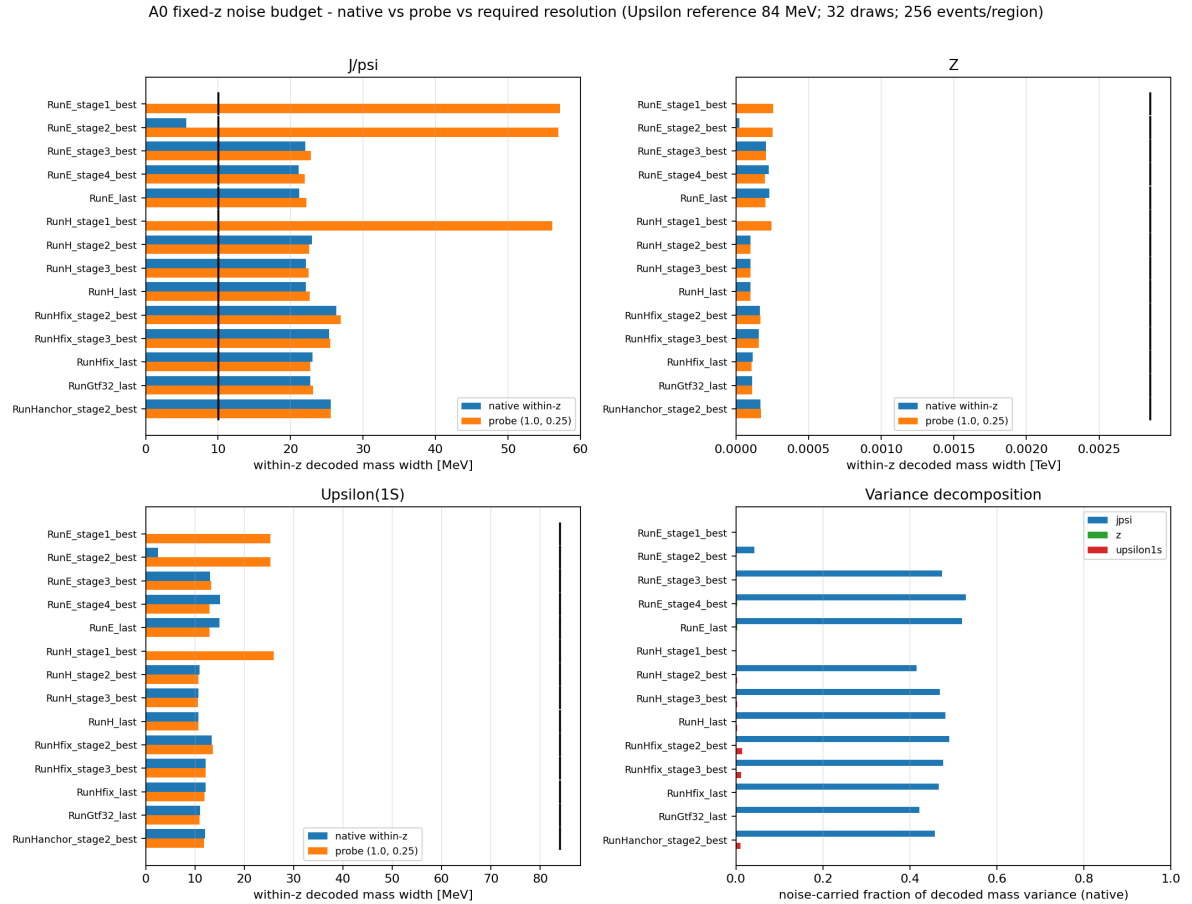


Figure 7: **Fixed-truth noise budget across arms.** Intra-event width against the resolution the detector requires, and the fraction of the variance carried by the noise, for 14 retained checkpoints spanning the Run E to Run H families. [measured]